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Peg hook

Abstract

A peg hook includes a mount and a rod extending from the mount. A tip is pivotally mounted to a distal end of the rod, the tip having a generally straight end and an arcuate body, wherein the tip and the rod each include a flat portion that are pinned to each other to permit the tip to rotate between an extended position and a retracted position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3481482	12/1968	Wilkins	N/A	N/A
3827569	12/1973	Canning	N/A	N/A
5711432	12/1997	Stein et al.	N/A	N/A
6481677	12/2001	Yu	84/327	A47F 5/083
7484628	12/2008	Schneider	N/A	N/A
7768399	12/2009	Hachmann et al.	N/A	N/A
8739981	12/2013	Piskor	N/A	N/A
11776517	12/2022	Gupta	84/453	F16M 11/2014
2005/0029208	12/2004	Paiste	84/421	A47F 7/00
2010/0032535	12/2009	Elliott	248/229.22	B25H 3/006

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This invention relates generally to a peg hook and, more particularly, to a theft deterrent peg hook to hinder removal of multiple items simultaneously.

Description of Prior Art

(2) A growing problem for retail stores is theft of high value inventory directly from shelves. Such shelves may include traditional store shelves stocked with multiple products or peg hooks which are likewise stocked with hanging rows of product. Thieves target retail stores to “sweep” shelves of valuable product by quickly assembling and removing as much of the product as possible. They then exit the store with the product before store personnel or law enforcement have an opportunity to react. The product is either later returned for cash or sold outright.

(3) Solutions to this problem have included locking down high value product in vaults or cabinets that require store personnel to unlock and physically provide the product to the consumer. Other solutions include security wires that physically connect the product to store shelves or each other. These solutions also require store personnel to unlock and untether such product for the consumer. These solutions tend to discourage customers from purchases because of the extra effort and time required to examiner and attain a product for purchase.

(4) Other solutions utilize mechanical dispensers connected to shelves that are configured to mechanically dispense one product at a time before resetting for dispensing another product. These solutions do not provide open access to the product to consumers and are not flexible for a variety of inventory.

(5) A need therefore exists in a retail system to slow suspected thieves down, maintain inventory while also enabling shoppers to examine and purchase individual products unencumbered by lock and key solutions.

SUMMARY OF THE INVENTION

(6) The invention generally relates to a theft deterrent system that can be used on a retail shelf peg hooks to protect certain goods from theft.

(7) The general object of the invention can be attained, at least in part, through a theft deterrent peg hook that includes a mount and a rod extending from the mount. A tip is pivotally affixed to a distal end of the rod and is moveable between an extended position and a retracted position.

(8) When a user removes a product from the peg hook, the tip preferably pivots between the extended position and the retracted position as the product is removed. Such pivoting action thereby prevents multiple products from being removed simultaneously. The throttling of product removal thereby mitigates the potential for a thief to “sweep” the peg hook, i.e., remove multiple or all products at once.

(9) Other objects and advantages will be apparent to those skilled in the art from the following detailed description taken in conjunction with the appended claims and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front side perspective view of a peg hook, in an extended position, according to one embodiment of the invention;

(2) FIG. 2 is a rear side perspective view of the peg hook shown in FIG. 1, in the extended position;

(3) FIG. 3 is a right side view of the peg hook shown in FIG. 1;

(4) FIG. 4 is a left side view, opposite the right side view shown in FIG. 3, of the peg hook shown in FIG. 1;

(5) FIG. 5 is a right side view of the peg hook shown in FIG. 1, in a retracted position; and

(6) FIG. 6 is a left side view, opposite the right side view shown in FIG. 5, of the peg hook shown in FIG. 1, in the retracted position

(7) FIG. 7 is a top view of the peg hook shown in FIG. 1;

(8) FIG. 8 is a side view of a peg hook according to one embodiment of the invention;

(9) FIG. 9 is a top side perspective view of the peg hook shown in FIG. 8;

(10) FIG. 10 is a top side perspective view of a peg hook according to one embodiment of the invention; and

(11) FIG. 11 is a front view of a peg hook according to one embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(12) FIGS. 1-11 show peg hooks 10 that can be mounted relative to a retail shelf (not shown) or similar retail fixture.

(13) The peg hook 10 preferably includes a mount 20. The mount 20 may comprise a mounting plate, such as shown in the figures; one, two or more prongs for engaging with a peg board; or other suitable mount 20 for attaching the peg hook 10 to the desired fixture, such as a store shelf, an endcap, or similar retail display.

(14) A rod 30 extends from the mount 20. The rod 30 includes a proximal end affixed to the mount 20 and a distal end. The rod 30 is preferably generally straight but may include an elevated portion at the distal end. In this way, products may be hung from the rod 30 and not slide off the end. FIG.

1 shows a product in phantom. Such products typically include a carded header through which an aperture is formed to hang the product from the peg hook **10**. Conventional peg hooks are simple rods that extend from a fixture upon which a column of products may be placed. These conventional peg hooks thereby permit a user to slide multiple products off the peg hook in a single action.

(15) As shown in the figures, a tip **50** is pivotally mounted to the distal end of the rod **30**. The tip **50** preferably includes a diameter approximately equal to the rod **30**. Together, the rod **30** and the tip **50** provide a generally continuous surface for hanging multiple hanging products from the peg hook **10**.

(16) The tip **50** preferably includes a generally straight end **60** and an arcuate body **70**. The tip **50** is preferably affixed to the rod **30** in a pivotable fashion such as through a pin **80**. The tip **50** preferably pivots between an extended position whereby the straight end **60** faces outward from the fixture and a retracted position whereby the straight end **60** is angled away from outward facing. FIGS. **3** and **4** best show the peg hook **10** in the extended position. FIGS. **5** and **6** best show the peg hook **10** in the retracted position.

(17) In one preferred embodiment, the tip **50** includes a flat portion **55** and the rod **30** includes a flat portion **40**. The respective flat portions **40**, **55** are preferably pinned to each other to create the pivoting action of the tip **50** relative to the rod **30**.

(18) In a preferred embodiment the flat portion **55** is a D cut and preferably comprises removal of approximately half of the rod **30**. Likewise, the flat portion **55** is approximately a half of the cross-section of the tip **50**.

(19) As shown in the figures, the tip **50** is connected to the rod **30** at their corresponding flat portions **40**, **55**. In this connection, the arcuate body **70** of the tip **50** preferably extends across the flat portion **40** of the rod **30**. The generally straight end **60** of the tip **50** is generally a consistent diameter and corresponds with the diameter of the majority of the length of the rod **30**.

(20) The tip **50** is preferably pinned to the rod **30** at a transition between the arcuate body **70** and the generally straight end **60**. A pin **80** may be fastened between the rod **30** and the tip **50** to enable the connection and rotation between the extended position and the retracted position. In this pinned arrangement, the tip **50** is configured to rotate between an extended position and a retracted position during removal of products from the peg hook **10**. Specifically, the tip **50** is preferably configured to rotate between an extended position wherein the arcuate body **70** extends above the rod **30** and a retracted position wherein the arcuate body **70** extends below the rod **30**. This extended position is best shown in FIGS. **3** and **4** and the retracted position is best shown in FIGS. **5** and **6**. FIGS. **8** and **9** illustrate both positions within the same drawings.

(21) In use, when a user removes a product from the peg hook **10**, the tip **50** preferably pivots between the extended position and the retracted position as the product is removed. Such pivoting action thereby prevents multiple products from being removed simultaneously as the arcuate body **70** and the straight end **60** of the tip **50** combine to block successive product from removal. The throttling of product removal thereby mitigates the potential for a thief to “sweep” the peg hook **10**, i.e., remove multiple or all products at once.

(22) The peg hook **10** is preferably configured so the tip **50** is elevated from the rod **30**. In addition, the distal end of the rod **30** is preferably elevated from a remaining length of the rod **30** and specifically the proximal end of the rod **30** at the mount **20**.

(23) According to an embodiment shown in FIGS. **8-10**, the peg hook **10** may further include a second rod **90** extending from the mount **20** generally parallel to the rod **30**. The second rod **90** may include a tag holder or similar plate **100** mounted to an end of the second rod **90**. The plate **100** is preferably positioned forward of the tip **50** and may accommodate product and/or pricing information.

(24) The invention illustratively disclosed herein suitably may be practiced in the absence of any element, part, step, component, or ingredient which is not specifically disclosed herein.

(25) While in the foregoing detailed description this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the invention is susceptible to additional embodiments and that certain of the details described herein can be varied considerably without departing from the basic principles of the invention.

Claims

1. A peg hook for a product having a header with an aperture configured to engage with the peg hook, the peg hook comprising: a mount; a rod extending from the mount, the rod having a diameter and a proximal end and a distal end; and a tip pivotally mounted to the distal end of the rod, the tip having a diameter approximately equal to the rod and a generally straight end and an arcuate body, wherein the tip and the rod each include a flat portion that are pinned to each other and wherein the rod and the tip provide a generally continuous surface for hanging multiple products from the peg hook, and wherein the arcuate body of the tip extends across the flat portion of the rod.
2. The peg hook of claim 1 wherein the flat portion is a D cut.
3. The peg hook of claim 2 wherein the D cut comprises approximately half of the rod.
4. The peg hook of claim 1 wherein the tip is pinned to the rod at a transition between the arcuate body and the generally straight end.
5. The peg hook of claim 1 wherein the tip is configured to rotate between an extended position and a retracted position.
6. The peg hook of claim 1 wherein the tip is elevated from the rod.
7. The peg hook of claim 1 wherein the distal end of the rod is elevated from a remaining length of the rod.
8. A peg hook for a product having a header with an aperture configured to engage with the peg hook, the peg hook comprising: a mount; a rod extending from the mount, the rod having a proximal end and a distal end; and a tip pivotally mounted to the distal end of the rod, the tip having a generally straight end and an arcuate body, wherein the tip and the rod each include a flat portion that are pinned to each other, wherein the arcuate body of the tip extends across the flat portion of the rod and wherein the tip is pinned to the rod at a transition between the arcuate body and the generally straight end and wherein the rod and the tip provide a generally continuous surface for hanging multiple products from the peg hook, and wherein the arcuate body of the tip extends across the flat portion of the rod.
9. The peg hook of claim 8 wherein the tip is elevated from the rod.
10. The peg hook of claim 8 wherein the distal end of the rod is elevated from a remaining length of the rod.
11. A peg hook for a product having a header with an aperture configured to engage with the peg hook, the peg hook comprising: a mount; a rod extending from the mount, the rod having a proximal end and a distal end; and a tip pivotally mounted to the distal end of the rod, the tip having a generally straight end and an arcuate body, wherein the tip and the rod each include a flat portion that are pinned to each other, wherein the tip is configured to rotate between an extended position wherein the arcuate body extends above the rod and a retracted position wherein the arcuate body extends below the rod and wherein the rod and the tip provide a generally continuous surface for hanging multiple products from the peg hook and wherein the arcuate body of the tip extends across the flat portion of the rod.
12. The peg hook of claim 11 wherein the tip is pinned to the rod at a transition between the arcuate body and the generally straight end.
13. The peg hook of claim 11 wherein the flat portion is approximately a half of a cross-section of

each of the rod and the tip.

14. The peg hook of claim 11 wherein the rod and the straight end of the tip are the same diameter.
